

Introduction & Motivation

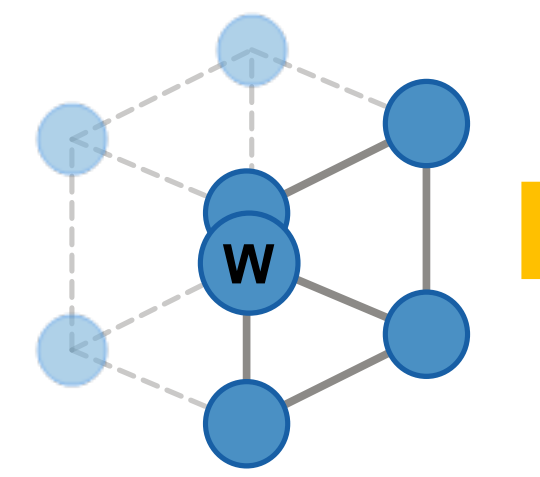
W is the highest melting point of all metals, ideal for plasma-facing components in fusion reactors

Challenge

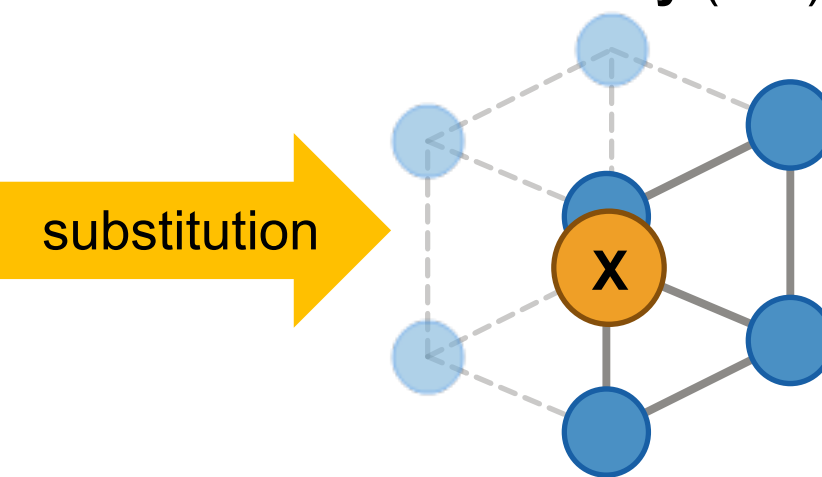
Pure W is **brittle** (low T)

DFT is **expensive**

Pure W system (bcc)



W-X alloy (bcc)



substitution

How can we efficiently screen elastic properties and thermal stability across W-Ta/Nb/Mo alloy compositions to identify optimal ductility-stability tradeoffs?

Goal

Systematically map elastic properties and thermal stability across W-based refractory alloys

UMA-OMAT *Universal Machine Learning Potential*

- Trained on Omat24 inorganic materials dataset
- DFT-level accuracy **~100× faster**

Elastic Properties

- C11, C12, C44 via energy-strain
- B, G via Voigt-Reuss-Hill
- Pugh's ratio $B/G > 1.75$
- Born stability criteria

Alloy Screening

- W-Ta / W-Nb / W-Mo
- 21 compositions each
- $x = 0 \rightarrow 1.00, \Delta x = 0.05$
- Tm via rule of mixtures

3 × 21 = 63
alloy systems compositions total calculations

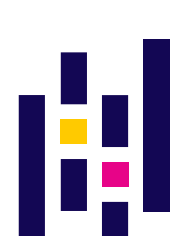
Data & Tools

Simulation



PyTorch

Analysis



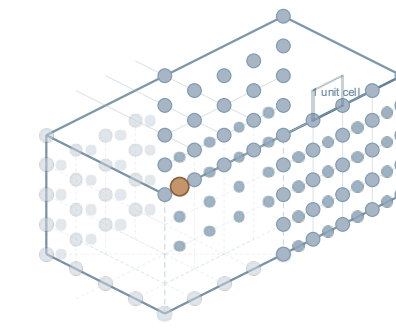
Data & Code



Methods & Workflow

Build Alloy

4x4x4 BCC supercell · 128 atoms
Hume-Rothery criterion satisfied: ($\Delta r < 5\%$)
Vegard's law: $a_0 = (1-x)a_W + x \cdot a_x$



Relax Structure & Compute Energy (UMA-OMAT)

FIRE optimizer: $f_{max} = 0.01 \text{ eV/\AA}$
Model: uma-s-1p1
Output: E_0

Deformation (Energy-Strain Method)

Hooke's Law:

$$\Delta E = \frac{V}{2} \sum_{i,j=1}^6 C_{ij} e_i e_j$$

$$\Delta E = E(\text{deformed}) - E_0$$

Cubic System: elastic constant matrix

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44} \end{bmatrix}$$

$$\frac{\Delta E}{V_0} = \frac{E - E_0}{V_0} = \frac{1}{2} C_{11} (e_1^2 + e_2^2 + e_3^2) + C_{12} (e_1 e_2 + e_2 e_3 + e_3 e_1) + \frac{1}{2} C_{44} (e_4^2 + e_5^2 + e_6^2) + O(\delta^3)$$

Hydrostatic

$$[F_{(hyd)}(\delta)] = \begin{bmatrix} 1+\delta & 0 & 0 \\ 0 & 1+\delta & 0 \\ 0 & 0 & 1+\delta \end{bmatrix}$$

$$e_1 = e_2 = e_3 = \delta$$

Shear

$$[F_{(shear)}(\delta)] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \delta/2 \\ 0 & \delta/2 & 1 \end{bmatrix}$$

$$e_4 = \delta, \quad e_1 = e_2 = e_3 = e_5 = e_6 = 0$$

Uniaxial Normal

$$[F_{(uni)}(\delta)] = \begin{bmatrix} 1+\delta & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$e_1 = \delta, \quad e_2 = e_3 = e_4 = e_5 = e_6 = 0$$

Compute Energy (UMA-OMAT)

$E = \text{atoms.get_potential_energy}()$
Record: deformation family, applied strain δ , $E(\text{deformed})$

Data Fitting and Extracting Elastic Constants

$$B = \frac{a_2}{4.5}$$

$$\frac{\Delta E}{V_0} = a_0 + a_2 \delta^2 + O(\delta^3)$$

$$C_{44} = 2a_2$$

$$C_{11} = 2a_2$$

$$C_{12} = \frac{3B - C_{11}}{2}$$

Calculate Macroscopic Polycrystalline Moduli

$$G = \frac{G_V + G_R}{2}$$

$$G_R = \frac{5(C_{11} - C_{12})C_{44}}{4C_{44} + 3(C_{11} - C_{12})}$$

$$G_V = \frac{C_{11} - C_{12} + 3C_{44}}{5}$$

Born Stability & Pugh Ratio & Melting Point

$$C_{11} - C_{12} > 0 \quad C_{11} + 2C_{12} > 0 \quad C_{44} > 0$$

$$\frac{B}{G} > 1.75$$

$$T_m = (1-x)T_m^W + x \cdot T_m^X$$

ASE

Torch
Sim

ASE

Results

Curve Fit

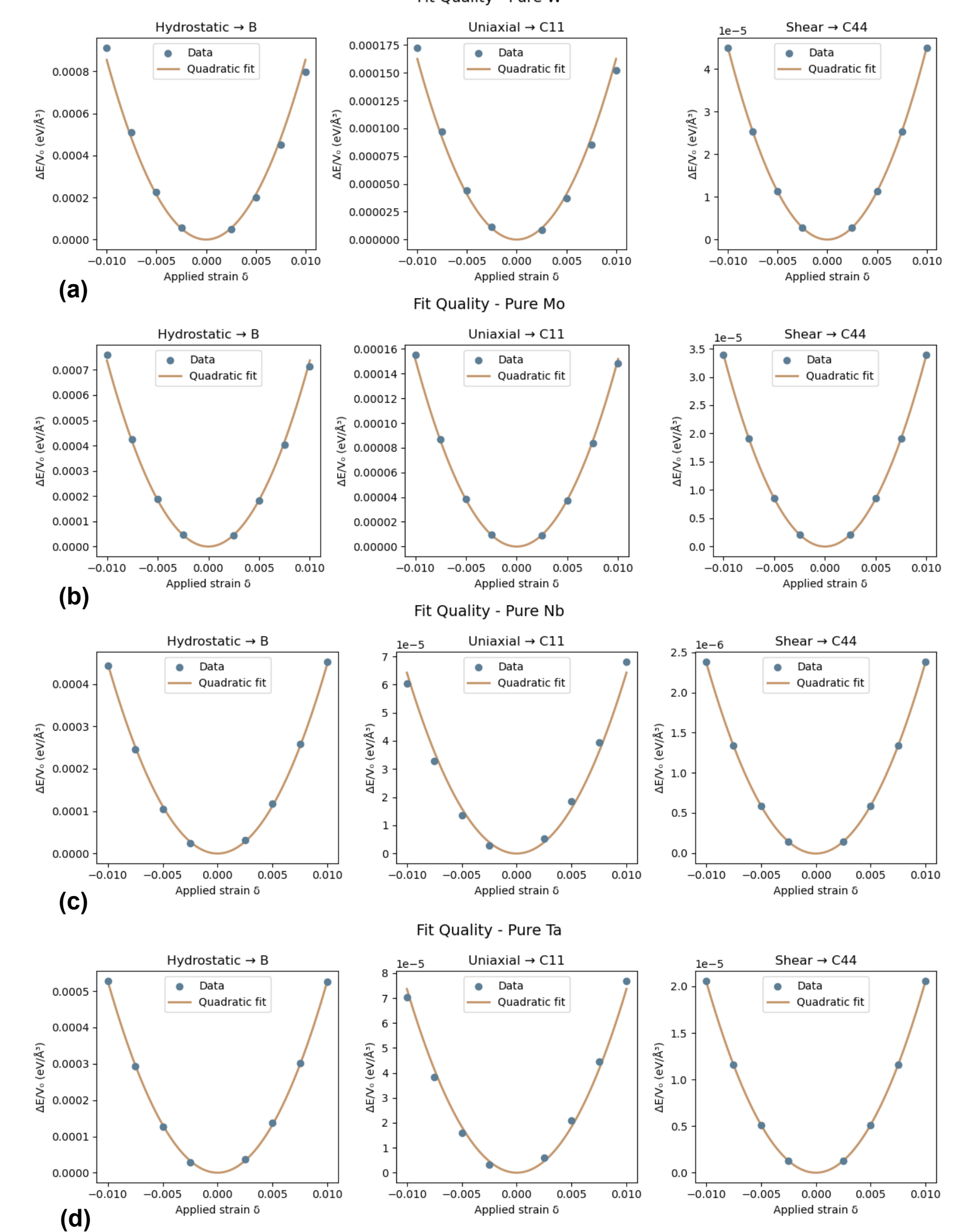


Figure 1. Validation of the energy-strain method for elastic constant extraction in pure bcc metals.

Comparison of B

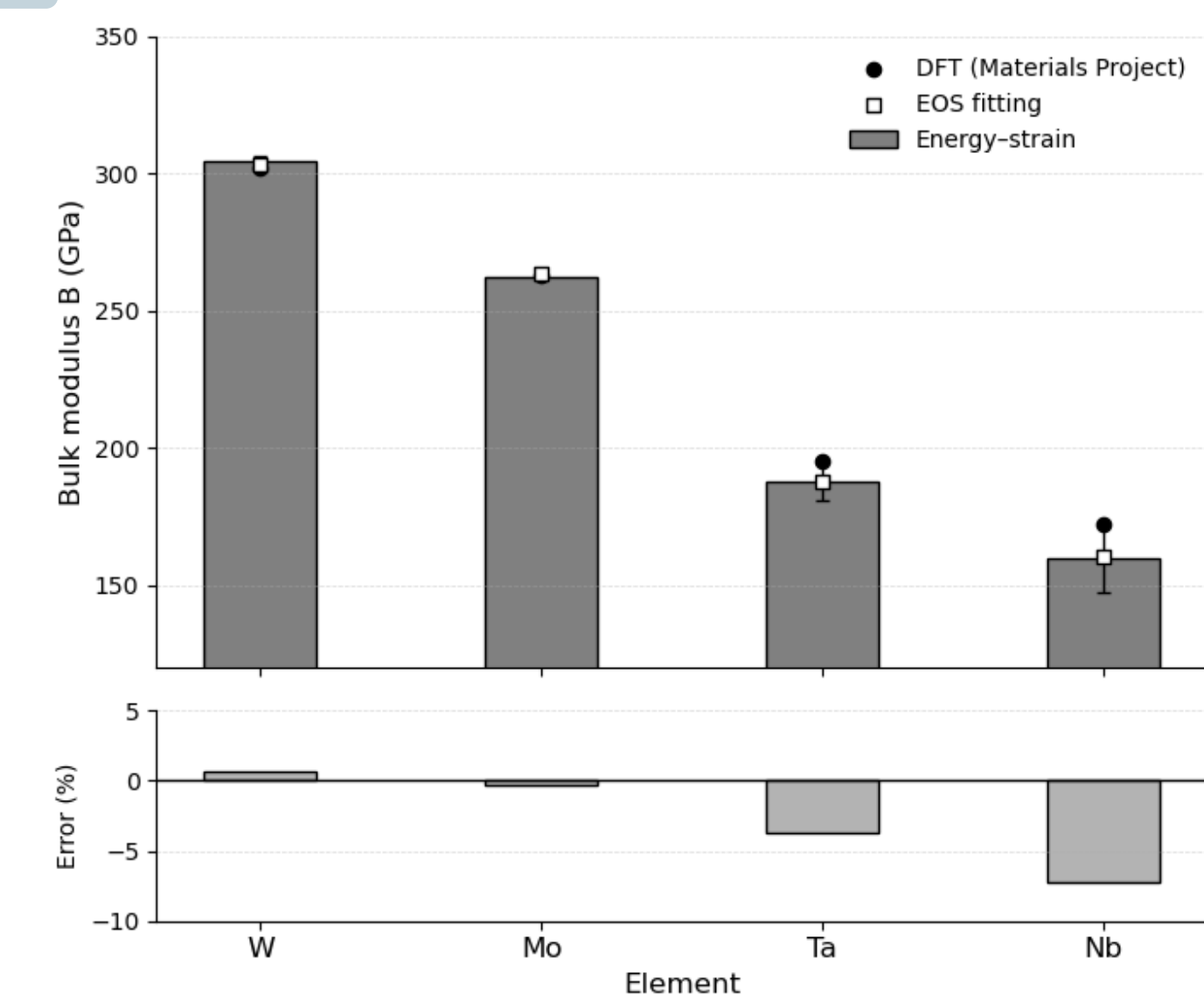


Figure 2. Validation of bulk modulus from energy-strain method against DFT reference

Future Work

Investigate G discrepancy in pure systems

- Test alternative shear deformation protocols
- Compare with stress-strain method as cross-validation

Extend to alloy system

- Apply validated methodology to full W-Ta/Nb/Mo alloy composition screening

Refine compositional resolution

- Reduce composition step from $\Delta x = 0.05$ to $\Delta x = 0.01$ for targeted composition ranges of interest

Reference

- [1] Landau, L.D.; Lifshitz, E.M. "Theory of Elasticity", 3rd ed.; Pergamon Press, 1986.
- [2] Teodosiu, C. "Elastic Models of Crystal Defects", Springer, 1982.
- [3] Mouhat, F.; Coudert, F.-X. Necessary and sufficient elastic stability conditions in various crystal systems. "Phys. Rev. B" 2014, 90, 224104.
- [4] Materials Project. <https://materialsproject.org>
- [5] Gangan, A. S.; Cohen, O. A.; Reibesehl, J.; et al. TorchSim: Simulation framework for atomistic modeling. <https://github.com/torchsim/torch-sim>